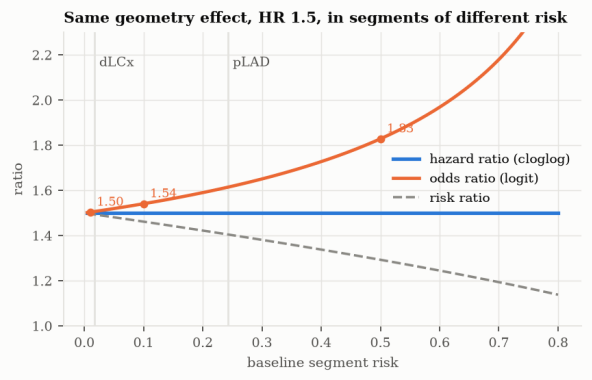
The pooled segment cloglog GLMM

Both the geometry and the outcome are recorded per segment: curvature and taper are measured in each of the nine trunk segments, and each segment is read for incident coronary plaque at follow-up. The risk model is therefore fitted on the segments, one row per segment. The who question is about the person, so the segment risks are then pooled into one person risk. The model is a cloglog generalized linear mixed model (GLMM): a cloglog link, a linear predictor, and a person random intercept. A person-level model of any plaque is kept as the benchmark. The figures use made-up numbers.

The cloglog link. Each segment is plaque-free at baseline and seen again about ten years later, and when plaque started is unknown. This is grouped survival data, for which the proportional hazards model is a cloglog model (Prentice and Gloeckler 1978). The name lists the steps: the *complement* $1-p$ is the probability that the segment stays plaque-free, the first *log* gives the hazard built up over follow-up, $\Lambda = -\log(1-p)$, and the second *log* gives $\eta = \log \Lambda$, which can take any value and is modelled as a sum of terms:

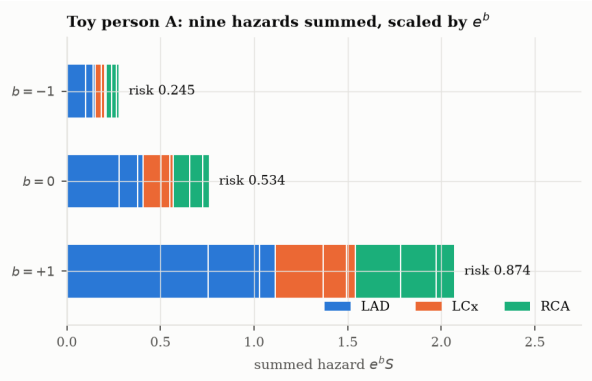
$$\operatorname{cloglog} P(y_{ij} = 1 \mid b_i) = \alpha_j + \beta_{\mathrm{S}}\,\mathrm{SCORE2}_i + \beta_{\mathrm{L}} \log \ell_{ij} + \mathrm{geometry}_{ij} + b_i,$$

with ℓ_{ij} the segment length. e^{β} is then a hazard ratio, which multiplies the hazard by the same factor in the pLAD and the rarer dLCx. An odds ratio would drift with the segment's baseline risk.



A hazard ratio of 1.5 is the same at every baseline risk; the odds ratio is not.

Hazards add, the person effect multiplies. A person is free of plaque only if all nine segments are, so the probabilities $e^{-\Lambda_j}$ multiply and the hazards add. The person risk depends on the segments only through the summed hazard $S_i = \sum_j e^{\eta_{ij}}$, and an unevaluable segment drops out of the sum. Risk that nobody measured acts on all segments of a person, so the nine rows of one person are not independent. The random intercept $b_i \sim N(0, \sigma^2)$ captures this: it multiplies all nine hazards of person i by e^{b_i}, and only σ is estimated.

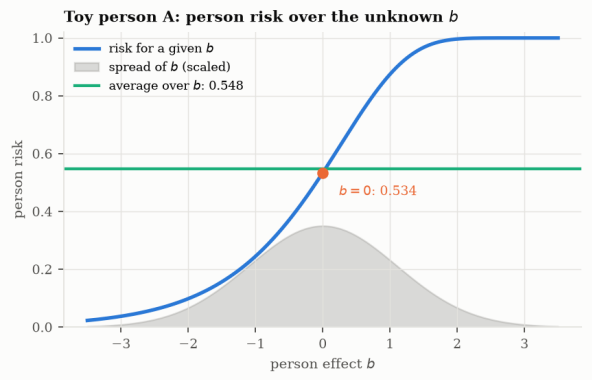


The nine segment hazards of one made-up person, stacked, for three values of the person effect b.

One person risk. For a new person b is unknown, so the person risk averages over it:

$$P_i = 1 - \int \exp\bigl(-e^{b} S_i\bigr)\, \phi(b; 0, \sigma^2)\, \mathrm{d}b,$$

computed by Gauss–Hermite quadrature. For the made-up person $P = 0.548$. Setting $b = 0$ gives 0.534, the risk of the median person, and multiplying the average segment risks, which ignores the shared b, gives 0.687. Both shortcuts rank people correctly, so only calibration detects them.



Person risk for each value of b (blue), the spread of b (grey) and the average used (green).

Fitting and testing. The weights and σ are fitted by maximum likelihood with b integrated out. Cross-validation is split by person, and P_i uses the σ of the training folds. Geometry is tested once, by a likelihood-ratio test of the geometry block; the gain in C and the calibration are estimated, not tested. The current code (`progression_who_pooled.py`) runs on the Medis trees with progression as the outcome and two geometry terms (mean turning and the LM angle).